

Cardiovascular system and Exercise

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Components of the Circulatory System

- ▶ **Heart**
- ▶ **Blood**
- ▶ **Blood vessels**



Why Do You Need the Circulatory System?

- ▶ Delivers **food and oxygen** to body cells
- ▶ Disposes of **carbon dioxide** and other waste products away from cells
- ▶ Helps to maintain a **constant body temperature**
- ▶ Aids the body in **fighting disease**



Heart

- ▶ Located near the centre of the chest
- ▶ Hollow structure
- ▶ About the size of your fist
- ▶ Made up of specialised muscle (**Cardiac Muscle**)
- ▶ Enclosed in a sac called **Pericardium**
- ▶ Three layers: Endocardium, Myocardium, Pericardium
- ▶ Myocardium – Contraction like a pump



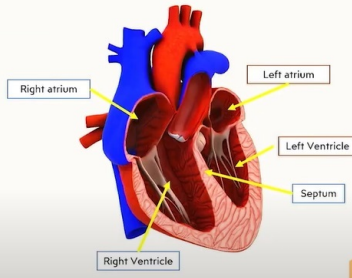
Heart - Chambers

- ▶ **4 chambers** in total
- ▶ **2 Atria (Upper chambers):**
 - ▶ Right Atrium: Receives Deoxygenated blood from the body
 - ▶ Left Atrium: Receives oxygenated blood from the lungs
- ▶ **2 Ventricles (Lower chambers):**
 - ▶ Right Ventricle: Pumps Deoxygenated blood to the lungs
 - ▶ Left Ventricle: Pumps oxygenated blood to the body
- ▶ **Septum**



Chambers of the Heart

Heart - Chambers



Heart Valves

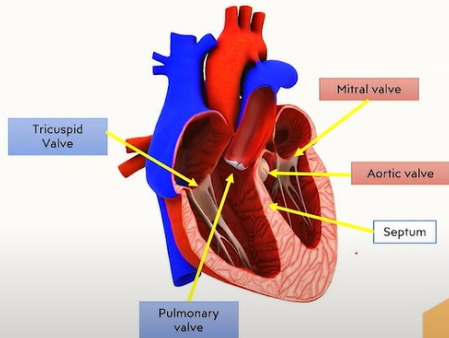
- ▶ **Tricuspid Valve:** Located between the right atrium and right ventricle, it prevents the backflow of blood as it moves from the right atrium to the right ventricle.
- ▶ **Pulmonary Valve:** Situated between the right ventricle and the pulmonary artery, it opens to allow deoxygenated blood to flow from the right ventricle to the lungs for oxygenation and closes to prevent backflow.
- ▶ **Mitral Valve:** Found between the left atrium and left ventricle, it ensures that oxygenated blood flows from the left atrium to the left ventricle without flowing backward.
- ▶ **Aortic Valve:** Located between the left ventricle and the aorta, it opens to allow oxygenated blood to flow from the left ventricle into the aorta for distribution to the rest of the body and closes to prevent backflow into the heart.



Heart valves

Heart - Valves

Right heart vs Left heart

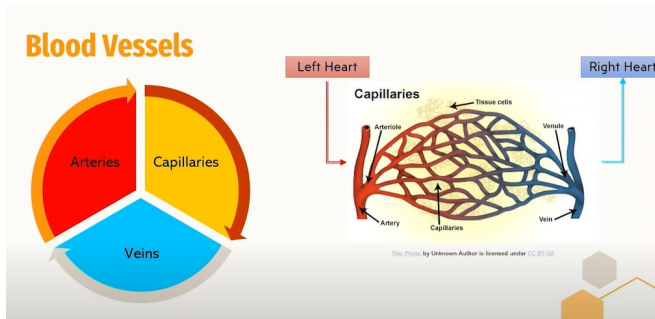


Blood Vessels

- ▶ **Arteries:** Carry oxygen-rich blood away from the heart to the body. They have thick, elastic walls to withstand high pressure. **Carries about 13% of blood at rest**
- ▶ **Capillaries:** The smallest blood vessels, connecting arteries to veins. Thin walls allow for the exchange of oxygen, nutrients, and waste between blood and tissues. **Small blood vessels contains about 7% of blood at rest** . Walls are just one cell thick & helps in the easy exchange of O₂ , CO₂, nutrients etc
- ▶ **Veins:** Carry oxygen-depleted blood back to the heart. Large blood vessels , **contains about 64% of blood at rest** .Large veins are situated near the skeletal muscles. Veins have valves that prevent back flow of blood Carry blood to heart from the different regions of the body .



Blood Vessels



Blood Composition

▶ **Plasma (55%):**

- ▶ Liquid component of blood, mostly water (about 90%)
- ▶ Contains proteins, electrolytes, hormones, and waste products
- ▶ Transports nutrients, hormones, and waste products throughout the body

▶ **Blood Cells (45%):**

- ▶ **Red Blood Cells (Erythrocytes):** Transport oxygen and carbon dioxide
- ▶ **White Blood Cells (Leukocytes):** Help fight infections
- ▶ **Platelets (Thrombocytes):** Involved in blood clotting and wound healing



Cardiovascular System (CVS) Parameters

- ▶ **Heart Rate:** Number of contractions per minute.
- ▶ **Stroke Volume:** The volume of blood pumped with each contraction (approximately 60 to 70 mL of blood).
- ▶ **Cardiac Output:** The total volume of blood pumped per minute (about 5 liters per minute).
- ▶ **Blood Pressure:** The pressure exerted by blood on the walls of the arteries.
- ▶ **Blood Distribution**



Blood Distribution

- ▶ **Metabolically active tissue** receives more blood supply due to higher oxygen and nutrient demands.
- ▶ **Intrinsic Regulation (Local Control):**
 - ▶ Local chemical factors such as low oxygen (O_2), high carbon dioxide (CO_2), and metabolic byproducts (e.g., lactic acid) cause vasodilation and increase blood flow.
 - ▶ Example: During exercise, working muscles produce more CO and lactic acid, which causes local blood vessels to dilate, increasing blood supply to those muscles.



Blood Distribution

▶ **Extrinsic Regulation (Systemic Control):**

- ▶ Controlled by the ****sympathetic nervous system****.
- ▶ Systemic control through the autonomic nervous system (particularly the sympathetic nervous system) and hormones, directing blood flow to key areas during stress, rest, or exercise.
- ▶ Example: During exercise, the sympathetic nervous system increases, causing vasoconstriction in non-active tissues (like the digestive system) and vasodilation in active muscles to optimize performance.



Heart Rate

▶ Resting Heart Rate:

- ▶ Normal range: 60 - 100 beats per minute (bpm)
- ▶ Less than 60 bpm: **Bradycardia**
- ▶ Greater than 100 bpm: **Tachycardia**

▶ Athlete's Bradycardia:

- ▶ Common in well-trained athletes
- ▶ Resting heart rate often below 60 bpm
- ▶ A sign of efficient heart function and cardiovascular fitness
- ▶ Typically not a concern unless accompanied by symptoms

▶ Measuring Resting Heart Rate:

- ▶ Early morning, before getting up from bed
- ▶ Measure for one full minute



Maximal and Submaximal Heart Rate

▶ Maximal Heart Rate (MHR):

- ▶ Maximum heart rate reached during maximal exertion or exercise
- ▶ Reduces with age
- ▶ Formula: **Maximal Heart Rate = 220 - Age**

▶ Submaximal Heart Rate (Submax HR):

- ▶ Heart rate during submaximal exercise (55-70% of MHR)
- ▶ Reached when exercise demand is primarily met through oxidative (aerobic) metabolism
- ▶ **Steady-State HR:** Submaximal HR stabilizes after a few minutes if exercise intensity remains constant
- ▶ Indicates how efficiently the cardiovascular system meets energy demands at moderate intensities
- ▶ Used to assess endurance and cardiovascular fitness



Heart Rate Reserve and Karvonen Formula

- ▶ **Heart Rate Reserve (HRR):**

- ▶ $HRR = \text{Maximum Heart Rate (MHR)} - \text{Resting Heart Rate (RHR)}$
- ▶ Reflects the available range within which the heart can operate during exercise

- ▶ **Karvonen Formula:**

- ▶ Target Heart Rate (THR) is calculated using:

$$THR = [(MHR - RHR) \times \text{Training Intensity \%}] + RHR$$

- ▶ Used to determine exercise intensity zones



Example: Target Heart rate (Karvonen Formula)

► **Given:**

- Resting Heart Rate (RHR) = 70 bpm
- Maximum Heart Rate (MHR) = 200 bpm
- Training Intensity = 60%

► **Step 1: Calculate HRR (Heart Rate Reserve):**

$$HRR = MHR - RHR = 200 - 70 = 130 \text{ bpm}$$

► **Step 2: Calculate THR (Target Heart Rate):**

$$THR = (HRR \times \text{Intensity}) + RHR$$

$$THR = (130 \times 0.6) + 70 = 78 + 70 = 148 \text{ bpm}$$

- **Conclusion:** The Target Heart Rate at 60% intensity is **148 bpm**.



Recovery Heart Rate

- ▶ **Recovery Heart Rate (RHR):**

- ▶ Heart rate dynamics post-exercise during the recovery period
- ▶ After exercise, heart rate gradually returns to the resting level

- ▶ **A Measure of Cardiorespiratory Fitness:**

- ▶ Rapid recovery indicates better cardiorespiratory fitness
- ▶ Normal recovery: Heart rate returns to resting levels over time

- ▶ **Significance:**

- ▶ **Failure of heart rate to decrease by more than 12 bpm in the first minute after peak exercise is a predictor of increased mortality risk**, especially in patients
- ▶ Used as a clinical marker for assessing overall heart health and fitness levels



Simple Test to Determine Recovery Heart Rate

► **2-Minute Step Test:**

- Perform moderate-intensity exercise (e.g., jogging or step-ups)
- Measure your heart rate immediately after the exercise
- Rest for 2 minutes, then measure your heart rate again
- The larger the difference between the initial heart rate and the heart rate after 2 minutes, the better the recovery

► **Interpretation:**

- A faster drop in heart rate during the 2-minute rest period indicates better cardiorespiratory fitness
- Slower recovery may indicate lower fitness or potential cardiovascular issues



Key Terms in Cardiovascular Physiology

- ▶ **End Systolic Volume (ESV):** The volume of blood remaining in the left ventricle at the end of systole (after contraction).
- ▶ **End Diastolic Volume (EDV):** The volume of blood in the left ventricle at the end of diastole (after filling).
- ▶ **Venous Return:** The flow of blood back to the heart from the veins, entering the right atrium.
- ▶ **Vasomotor Tone:** The level of constriction in blood vessels, primarily arterioles, regulated by the autonomic nervous system to control blood pressure and flow.



Exercise and Stroke Volume

Stroke Volume (SV) is the amount of blood pumped by the heart per beat. During exercise, stroke volume increases to meet higher oxygen demands:

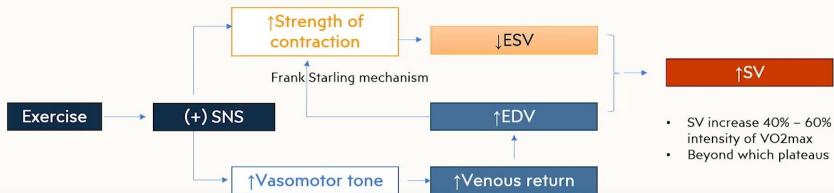
- ▶ **Increased venous return:** More blood flows back to the heart, stretching the heart and causing stronger contractions.
- ▶ **Increased contractility:** The heart beats more forcefully due to stimulation from the sympathetic nervous system.

Regular exercise enhances SV by improving heart strength and efficiency, enabling athletes to pump more blood per beat and improve cardiovascular endurance.



Exercise and Stroke Volume

Exercise & Stroke Volume



Frank-Starling mechanism: The stretched fibers result in a stronger contraction, increasing stroke volume.



Exercise and Cardiac Output

Cardiac Output (CO) is the total volume of blood the heart pumps per minute. It is a product of heart rate (HR) and stroke volume (SV):

$$CO = HR \times SV$$

During exercise, cardiac output increases significantly to supply more oxygen to working muscles:

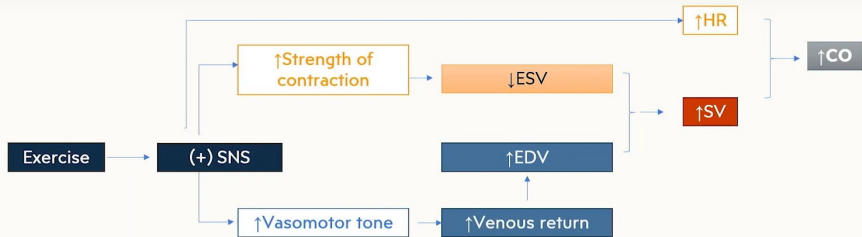
- ▶ **Increased Heart Rate:** The heart beats faster to pump more blood.
- ▶ **Increased Stroke Volume:** More blood is pumped per beat due to stronger heart contractions.



Exercise and Cardiac Output

Regular exercise improves cardiac output by enhancing both heart rate and stroke volume, which increases the efficiency of oxygen delivery to tissues.

Exercise & Cardiac Output

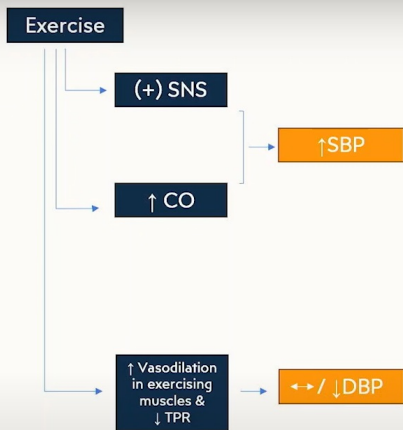


- Normal values
 - Resting ~5 L/min
 - Untrained ~20 L/min
 - Trained ~ 40 L/min



Exercise and Blood Pressure

Exercise & Blood Pressure



Total Peripheral Resistance (TPR)



Risks of Valsalva Maneuver in Physical Activity

Risks of Valsalva Maneuver

- Reduced Blood Flow to the Brain: Drop in venous return can reduce blood flow to the brain, potentially causing dizziness or even fainting (syncope) during heavy lifting.
- Excessive Blood Pressure Spikes: Valsalva maneuver can cause dangerously high spikes in blood pressure, increasing the risk of stroke or other cardiovascular events.

Proper Breathing Technique

To minimize risks, athletes are often advised to practice proper breathing techniques during strength training. "Exhaling during the exertion phase" of the lift (rather than holding the breath) helps prevent excessive intrathoracic pressure and reduces cardiovascular strain.



Exercise and Blood Flow Distribution

Exercise & Blood flow distribution



Liver, Stomach & Gut

20-25%

3-5%



Brain

15%

3-4%



Kidney

20%

2-4%



Skeletal Muscle

15-20%

70-85%



Heart

4-5%

4-5%



Skin

4-5%

5 - 20%



Bones

3-5%



Athlete's Heart:

Key Characteristics of Athlete's Heart:

- ▶ **Enlarged Heart Chambers:** The left ventricle, often becomes larger,
- ▶ **Lower Resting Heart Rate:** Athletes typically have a lower resting heart rate (bradycardia).
- ▶ **Increased Stroke Volume:** The amount of blood pumped with each heartbeat (stroke volume) increases, meaning the heart can work less to circulate blood.
- ▶ **Normal Blood Pressure:** Despite the heart enlargement, blood pressure typically remains normal or even lower due to better vascular function.



Hypertrophy in Endurance vs. Strength Athletes

Key Differences

Eccentric Hypertrophy (Endurance Athletes)

- ▶ Ventricular chamber enlarges
- ▶ Wall thickens proportionally
- ▶ Increased stroke volume
- ▶ Enhanced endurance capacity

Concentric Hypertrophy (Strength Athletes)

- ▶ Ventricular wall thickens without chamber enlargement
- ▶ Handles increased pressure during short bursts



Blood Flow Adaptations due to Exercise

- ▶ With increasing demand, **blood flow to exercising muscles** increases.
- ▶ **Capillarization** improves:
 - ▶ Increase in the **number of capillaries**.
 - ▶ Recruitment of **existing capillaries** to supply more blood.
- ▶ **Redistribution of blood flow** from intra-abdominal organs to working muscles becomes more effective.



Erythropoietin and RBC Volume

- ▶ Aerobic training stimulates the hormone **erythropoietin**, which increases **red blood cell (RBC) volume**.
- ▶ More RBCs mean more **hemoglobin**, enhancing oxygen-carrying capacity for better endurance.



